

2.4 解:

$$\theta_1 = 30^\circ \quad \theta_2 = 0^\circ \quad \theta_3 = 45^\circ$$

$$R_z(\theta_1) = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_y(\theta_2) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_x(\theta_3) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_3 & -\sin\theta_3 \\ 0 & \sin\theta_3 & \cos\theta_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$R = R_z(\theta_1) R_y(\theta_2) R_x(\theta_3) = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}}{4} \\ \frac{1}{2} & \frac{\sqrt{6}}{4} & -\frac{\sqrt{6}}{4} \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

2.13 解:

$$\begin{bmatrix} B \\ C \end{bmatrix}^T = \begin{bmatrix} A & B \\ C & A \end{bmatrix}^T \begin{bmatrix} B^T & A^T \\ A^T & C^T \end{bmatrix} = \begin{bmatrix} B^T & A^T \\ A^T & C^T \end{bmatrix} \begin{bmatrix} A & B \\ C & A \end{bmatrix}$$

$$\begin{bmatrix} A \\ C \end{bmatrix}^T = \begin{bmatrix} U^T & V^T \\ A^T & C^T \end{bmatrix} \begin{bmatrix} 0.866 & -0.5 & 0 & 1 \\ 0.5 & 0.866 & 0 & -1 \\ 0 & 0 & 1 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.866 & -0.5 & 0 & -3 \\ 0.433 & 0.75 & -0.5 & -3 \\ 0.25 & 0.433 & 0.866 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}^{-1}$$

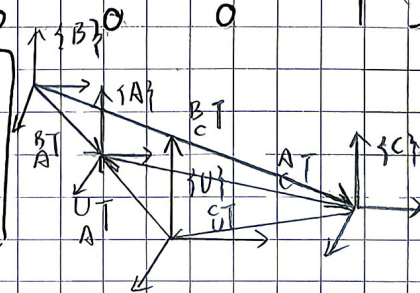
$$\begin{bmatrix} 0.866 & -0.5 & 0 & 1 \\ 0.5 & 0.866 & 0 & -1 \\ 0 & 0 & 1 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.866 & -0.5 & 0 & -3 \\ 0.433 & 0.75 & -0.5 & -3 \\ 0.25 & 0.433 & 0.866 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}^{-1}$$

$$\begin{bmatrix} 0.866 & -0.5 & 0 & -9.026 \\ -0.5 & 0.866 & 0 & 6.366 \\ 0 & 0 & 1 & -8 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.866 & 0.433 & 0.25 & 3.147 \\ -0.5 & 0.75 & 0.433 & -0.55 \\ 0 & -0.5 & 0.866 & -4.1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0.5 & 0.75 & 0.433 & -6.576 \\ -0.866 & 0.433 & 0.25 & 4.316 \\ 0 & -0.5 & 0.866 & -12.1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} B \\ C \end{bmatrix}^T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.866 & -0.5 & 10 \\ 0 & 0.5 & 0.866 & -20 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.5 & 0.75 & 0.433 & -6.576 \\ -0.866 & 0.433 & 0.25 & 4.316 \\ 0 & -0.5 & 0.866 & -12.1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 & 0.75 & 0.433 & -6.58 \\ -0.75 & 0.625 & -0.217 & 19.79 \\ -0.433 & -0.217 & 0.875 & -28.32 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



2.27

$$A_R = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A_{P_{\text{CORG}}} = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$$

$$A_T = \begin{bmatrix} -1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.29

$$A_R = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ 0 & -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ -1 & 0 & 0 \end{bmatrix}$$

$$A_{P_{\text{CORG}}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$A_{P_{\text{CORG}}} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$$A_T = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ 0 & -\frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.28

$$A_R = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$A_{P_{\text{CORG}}} = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$$

$$A_T = \begin{bmatrix} 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} & 3 \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.29 2.30

$$A_T = (A_R)^T =$$

$$A_T = (A_R)^T = \begin{bmatrix} A_R^T & -A_R^T A_{P_{\text{CORG}}} \\ 0 & \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 & 2 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 & \frac{3}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 & -\frac{\sqrt{3}}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2.31

$$A_R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$A_{P_{\text{CORG}}} = \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}$$

$$A_T = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 4 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 4 \\ 0 & -1 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.32

$$A_R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} \cos(-36.9^\circ) & -\sin(-36.9^\circ) & 0 \\ \sin(-36.9^\circ) & \cos(-36.9^\circ) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0.8 & 0.6 & 0 \\ -0.6 & 0.8 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.8 & 0.6 & 0 \\ 0.6 & -0.8 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$A_{P_{\text{CORG}}} = \begin{bmatrix} -3 \\ 4 \\ 2 \end{bmatrix}$$

$$A_T = \begin{bmatrix} 0.8 & 0.6 & 0 & -3 \\ 0.6 & -0.8 & 0 & 4 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2

2-33

$${}^B_c R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} -0.8 & 0.6 & 0 \\ -0.6 & -0.8 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -0.8 & 0.6 & 0 \\ 0 & 0 & 1 \\ 0.6 & 0.8 & 0 \end{bmatrix}$$

$${}^B P_{\text{CORG}} = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$$

$${}^B_c \bar{T} = \begin{bmatrix} -0.8 & 0.6 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ 0.6 & 0.8 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2-34

$${}^C_A \bar{T} = ({}^A_c T)^T = \begin{bmatrix} {}^A_c R^T & -{}^A_c R^T {}^A P_{\text{CORG}} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.8 & 0.6 & 0 & 0 \\ 0.6 & -0.8 & 0 & -3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$